CS-670 Course Project

Supplementary Results and Experimental Record

Privacy, Representation, and Noise Propagation in Federated Recommendation

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Full retained utility tables, statistical contrasts, geometry diagnostics, personalization ablations, and reproducibility information accompanying the main CS-670 project report. Tables are generated from frozen saved evidence.

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S1 How to interpret this supplement

The main report develops the research questions, methods, and conclusions. This document retains the numerical detail needed to inspect those conclusions and the unfavorable results. It is not an additional experiment or an independent set of replications. All numbers come from the saved source tables identified in the provenance record.

Privacy labels. Targets 8/4/2/1 refer to a single trained model at $\delta = 10^{-5}$. "No DP" means the matched no-clipping/no-noise control where defined; "Clip only" retains clipping but adds no Gaussian noise. Diagnostic continuation is not covered by the starting checkpoint's epsilon. Popularity uses a single full-population release and its own calibration.

Means and uncertainty. Utility entries show mean ± seed SD. SD is not a confidence interval. Paired bootstrap contrasts average each user's metric over five seeds before resampling users 100,000 times. Their ordinary and adjusted intervals therefore condition on the retained training runs. Primary families contain four B3 contrasts, sixteen B4 contrasts, and sixteen E1 FixedB-minus-Two contrasts. Other families are secondary. An interval including zero is not an equivalence finding.

Historical names. "B4 diag." is the historical fixed-setting diagnostic at common optimizer settings. It is not the E1 model with a fixed public factor. "Hist. B4" in E1 comparison rows identifies an archived historical comparator. This distinction matters because the naming predates the fixed-public-factor extension.

Scope of completeness. All retained main and secondary comparison rows are included. Geometry, activity, and personalization tables retain all retained settings. E2/E2b endpoints cover both declared amplitudes and all retained intervention branches; full time series, seed medians, and raw observations remain in the research archive. E3's table includes all 20 shared-reset endpoint settings. E4's ten screening cells cover all 50 states and 3,200 pairs. Bulk per-user records, checkpoints, and abandoned search logs are not duplicated in a printable appendix.

S2 Historical utility and paired comparisons

Table S1: Phase-0 random and raw-popularity references.

Model	Split	NDCG@10	User SE	HR@10	MRR@10
random	validation	0.002266	0.000942	0.006369	0.001085
random	test	0.003773	0.001491	0.008493	0.002410
popularity	validation	0.046378	0.005599	0.083864	0.035093
popularity	test	0.044292	0.005209	0.087049	0.031442

SE describes across-user metric uncertainty for the retained reference, not training-seed SD. Random ranking uses seed 42; raw popularity is nonprivate. Both use the frozen full-ranking evaluator.

Table S3: Three-seed weighting and federation decomposition.

Comparison	Mean difference	95% interval
B0-UW - B0	0.005773	[-0.000164, 0.011910]
B1 - B0-UW	-0.004206	[-0.010903, 0.002288]
B1 - B0	0.001567	[-0.005353, 0.008668]

Table S2: Matched centralized and federated controls (three seeds).

Model	NDCG@10	Seed SD	HR@10	MRR@10
B0	0.085972	0.000456	0.171975	0.060051
B0-UW	0.091745	0.001866	0.181882	0.064536
B1	0.087539	0.002074	0.164190	0.064527

Seeds 42/123/2026. This B1 mean differs from the five-seed B1 mean used in later comparisons. Source: centralized_federated_3seeds.csv.

These overall paired intervals all include zero. This family uses the matched three-seed control, not the later five-seed B1 mean.

Table S4: All retained five-seed historical utility rows.

Model	NDCG@10	HR@10	MRR@10
B1	0.08901 ± 0.00249	0.16709 ± 0.00555	0.06555 ± 0.00167
B2 nodp	0.05789 ± 0.00163	0.10934 ± 0.00360	0.04236 ± 0.00168
B2 $\epsilon = 8$	0.05227 ± 0.00321	0.09894 ± 0.00630	0.03810 ± 0.00237
B2 $\epsilon = 4$	0.04217 ± 0.00491	0.07856 ± 0.00819	0.03110 ± 0.00442
B2 $\epsilon = 2$	0.02543 ± 0.00316	0.04565 ± 0.00859	0.01934 ± 0.00186
B2 $\epsilon = 1$	0.01146 ± 0.00095	0.02166 ± 0.00178	0.00838 ± 0.00110
B3 r4	0.05947 ± 0.00381	0.11953 ± 0.00800	0.04154 ± 0.00281
B3 r8	0.07101 ± 0.00596	0.14501 ± 0.01470	0.04894 ± 0.00364
B3 r16	0.07633 ± 0.00722	0.15202 ± 0.01470	0.05366 ± 0.00530
B3 r32	0.06807 ± 0.01354	0.13206 ± 0.02665	0.04885 ± 0.00966
B4 diag. r4 $\epsilon = 4$	0.04583 ± 0.00339	0.08854 ± 0.00851	0.03291 ± 0.00198
B4 diag. r8 $\epsilon = 4$	0.04527 ± 0.00135	0.09045 ± 0.00625	0.03169 ± 0.00064
B4 diag. r16 $\epsilon = 4$	0.04347 ± 0.00567	0.08471 ± 0.00936	0.03096 ± 0.00473
B4 diag. r32 $\epsilon = 4$	0.04564 ± 0.00303	0.08832 ± 0.00774	0.03273 ± 0.00203
B4 r4 $\epsilon = 1$	0.01598 ± 0.00335	0.03333 ± 0.00707	0.01084 ± 0.00251
B4 r4 $\epsilon = 2$	0.02908 ± 0.00944	0.05499 ± 0.01616	0.02125 ± 0.00762
B4 r4 $\epsilon = 4$	0.03785 ± 0.00834	0.07176 ± 0.01600	0.02757 ± 0.00644
B4 r4 $\epsilon = 8$	0.03854 ± 0.00978	0.07389 ± 0.01866	0.02785 ± 0.00714
B4 r4 nodp	0.04083 ± 0.00538	0.07495 ± 0.01280	0.03039 ± 0.00349
B4 r8 $\epsilon = 1$	0.01816 ± 0.00462	0.03376 ± 0.00924	0.01348 ± 0.00347
B4 r8 $\epsilon = 2$	0.03449 ± 0.00336	0.06667 ± 0.00714	0.02489 ± 0.00222
B4 r8 $\epsilon = 4$	0.04265 ± 0.00181	0.08344 ± 0.00560	0.03035 ± 0.00104
B4 r8 $\epsilon = 8$	0.04539 ± 0.00507	0.08641 ± 0.01244	0.03296 ± 0.00321
B4 r8 nodp	0.04808 ± 0.00966	0.09278 ± 0.01554	0.03463 ± 0.00796
B4 r16 $\epsilon = 1$	0.01668 ± 0.00557	0.03185 ± 0.01121	0.01214 ± 0.00404
B4 r16 $\epsilon = 2$	0.03399 ± 0.00581	0.06433 ± 0.01172	0.02482 ± 0.00444
B4 r16 $\epsilon = 4$	0.03827 ± 0.00770	0.07049 ± 0.01373	0.02845 ± 0.00594
B4 r16 $\epsilon = 8$	0.04363 ± 0.00630	0.08408 ± 0.01209	0.03144 ± 0.00465
B4 r16 nodp	0.04914 ± 0.00516	0.09278 ± 0.00877	0.03600 ± 0.00422
B4 r32 $\epsilon = 1$	0.01539 ± 0.00310	0.02994 ± 0.00494	0.01097 ± 0.00251
B4 r32 $\epsilon = 2$	0.03650 ± 0.00536	0.06879 ± 0.01127	0.02676 ± 0.00403
B4 r32 $\epsilon = 4$	0.04449 ± 0.00590	0.08450 ± 0.01260	0.03232 ± 0.00400
B4 r32 $\epsilon = 8$	0.04402 ± 0.00531	0.08450 ± 0.00980	0.03187 ± 0.00438
B4 r32 nodp	0.05093 ± 0.00284	0.09957 ± 0.00414	0.03634 ± 0.00264

Entries are mean ± seed SD. B4 diag. denotes the historical fixed-setting diagnostic, not E1-FixedB.

Table S5: Historical paired-user contrasts, including secondary families.

Contrast ($a - b$)	Difference	95% interval	Adjusted interval	Decision
B3_vs_B1				
B3 r4 − B1	-0.029540	[-0.039095, -0.020300]	[-0.041779, -0.017893]	Negative
B3 r8 − B1	-0.017998	[-0.025420, -0.010877]	[-0.027511, -0.008949]	Negative
B3 r16 − B1	-0.012675	[-0.019215, -0.006262]	[-0.020950, -0.004560]	Negative
B3 r32 − B1	-0.020934	[-0.028896, -0.013228]	[-0.031093, -0.011160]	Negative
B4_vs_B2				
B4 r4 $\epsilon = 8$ − B2 $\epsilon = 8$	-0.013726	[-0.019314, -0.008312]	[-0.022317, -0.005598]	Negative
B4 r4 $\epsilon = 4$ − B2 $\epsilon = 4$	-0.004324	[-0.009148, 0.000522]	[-0.011585, 0.002992]	Includes zero
B4 r4 $\epsilon = 2$ − B2 $\epsilon = 2$	0.003651	[-0.000861, 0.008216]	[-0.003142, 0.010626]	Includes zero
B4 r4 $\epsilon = 1$ − B2 $\epsilon = 1$	0.004515	[0.000676, 0.008452]	[-0.001316, 0.010500]	Includes zero
B4 r8 $\epsilon = 8$ − B2 $\epsilon = 8$	-0.006886	[-0.011402, -0.002495]	[-0.013707, -0.000203]	Negative
B4 r8 $\epsilon = 4$ − B2 $\epsilon = 4$	0.000483	[-0.004269, 0.005293]	[-0.006698, 0.007825]	Includes zero
B4 r8 $\epsilon = 2$ − B2 $\epsilon = 2$	0.009057	[0.003971, 0.014214]	[0.001486, 0.016931]	Positive
B4 r8 $\epsilon = 1$ − B2 $\epsilon = 1$	0.006701	[0.002476, 0.011139]	[0.000495, 0.013578]	Positive
B4 r16 $\epsilon = 8$ − B2 $\epsilon = 8$	-0.008643	[-0.013185, -0.004219]	[-0.015488, -0.001965]	Negative
B4 r16 $\epsilon = 4$ − B2 $\epsilon = 4$	-0.003895	[-0.008587, 0.000832]	[-0.010921, 0.003248]	Includes zero
B4 r16 $\epsilon = 2$ − B2 $\epsilon = 2$	0.008565	[0.003478, 0.013806]	[0.001047, 0.016664]	Positive
B4 r16 $\epsilon = 1$ − B2 $\epsilon = 1$	0.005224	[0.001044, 0.009662]	[-0.000945, 0.012136]	Includes zero
B4 r32 $\epsilon = 8$ − B2 $\epsilon = 8$	-0.008249	[-0.012781, -0.003743]	[-0.015076, -0.001407]	Negative
B4 r32 $\epsilon = 4$ − B2 $\epsilon = 4$	0.002319	[-0.002590, 0.007299]	[-0.005037, 0.009856]	Includes zero
B4 r32 $\epsilon = 2$ − B2 $\epsilon = 2$	0.011075	[0.005667, 0.016690]	[0.003012, 0.019719]	Positive
B4 r32 $\epsilon = 1$ − B2 $\epsilon = 1$	0.003929	[0.000007, 0.007923]	[-0.001939, 0.009926]	Includes zero
sec_selected_rank				
B4 r32 nodp − B2 nodp	-0.006958	[-0.010735, -0.003333]	—	Secondary
B4 r8 $\epsilon = 8$ − B2 $\epsilon = 8$	-0.006886	[-0.011402, -0.002495]	—	Secondary
B4 r32 $\epsilon = 4$ − B2 $\epsilon = 4$	0.002319	[-0.002590, 0.007299]	—	Secondary
B4 r16 $\epsilon = 2$ − B2 $\epsilon = 2$	0.008565	[0.003478, 0.013806]	—	Secondary
B4 r16 $\epsilon = 1$ − B2 $\epsilon = 1$	0.005224	[0.001044, 0.009662]	—	Secondary
sec_nodp_capacity				
B4 r4 nodp − B2 nodp	-0.017055	[-0.023217, -0.011166]	—	Secondary
B4 r8 nodp − B2 nodp	-0.009806	[-0.014111, -0.005641]	—	Secondary
B4 r16 nodp − B2 nodp	-0.008745	[-0.012655, -0.005007]	—	Secondary
B4 r32 nodp − B2 nodp	-0.006958	[-0.010735, -0.003333]	—	Secondary
sec_own_dp_loss				
B4 r4 $\epsilon = 8$ − B4 r4 nodp	-0.002289	[-0.006012, 0.001423]	—	Secondary
B4 r4 $\epsilon = 4$ − B4 r4 nodp	-0.002988	[-0.007220, 0.001162]	—	Secondary
B4 r4 $\epsilon = 2$ − B4 r4 nodp	-0.011755	[-0.017990, -0.005790]	—	Secondary
B4 r4 $\epsilon = 1$ − B4 r4 nodp	-0.024858	[-0.032077, -0.017976]	—	Secondary
B4 r8 $\epsilon = 8$ − B4 r8 nodp	-0.002698	[-0.006732, 0.001256]	—	Secondary
B4 r8 $\epsilon = 4$ − B4 r8 nodp	-0.005430	[-0.009869, -0.001084]	—	Secondary
B4 r8 $\epsilon = 2$ − B4 r8 nodp	-0.013598	[-0.019846, -0.007513]	—	Secondary

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Table S5 (continued)

Contrast ($a - b$)	Difference	95% interval	Adjusted interval	Decision
B4 r8 $\epsilon = 1$ − B4 r8 nodp	-0.029920	[-0.037482, -0.022633]	—	Secondary
B4 r16 $\epsilon = 8$ − B4 r16 nodp	-0.005515	[-0.009723, -0.001426]	—	Secondary
B4 r16 $\epsilon = 4$ − B4 r16 nodp	-0.010869	[-0.015534, -0.006436]	—	Secondary
B4 r16 $\epsilon = 2$ − B4 r16 nodp	-0.015150	[-0.022212, -0.008351]	—	Secondary
B4 r16 $\epsilon = 1$ − B4 r16 nodp	-0.032459	[-0.041521, -0.023774]	—	Secondary
B4 r32 $\epsilon = 8$ − B4 r32 nodp	-0.006909	[-0.011453, -0.002508]	—	Secondary
B4 r32 $\epsilon = 4$ − B4 r32 nodp	-0.006442	[-0.012077, -0.000920]	—	Secondary
B4 r32 $\epsilon = 2$ − B4 r32 nodp	-0.014427	[-0.021491, -0.007607]	—	Secondary
B4 r32 $\epsilon = 1$ − B4 r32 nodp	-0.035541	[-0.044833, -0.026584]	—	Secondary
sec_fixed_diag_eps4				
B4 diag. r4 $\epsilon = 4$ − B2 $\epsilon = 4$	0.003658	[-0.001238, 0.008641]	—	Secondary
B4 diag. r8 $\epsilon = 4$ − B2 $\epsilon = 4$	0.003104	[-0.002063, 0.008370]	—	Secondary
B4 diag. r16 $\epsilon = 4$ − B2 $\epsilon = 4$	0.001305	[-0.003751, 0.006498]	—	Secondary
B4 diag. r32 $\epsilon = 4$ − B2 $\epsilon = 4$	0.003465	[-0.001692, 0.008727]	—	Secondary
ref_B2_dp_loss				
B2 $\epsilon = 8$ − B2 nodp	-0.005618	[-0.010669, -0.000715]	—	Secondary
B2 $\epsilon = 4$ − B2 nodp	-0.015719	[-0.022748, -0.009040]	—	Secondary
B2 $\epsilon = 2$ − B2 nodp	-0.032460	[-0.041925, -0.023421]	—	Secondary
B2 $\epsilon = 1$ − B2 nodp	-0.046428	[-0.057482, -0.035834]	—	Secondary

Decisions refer only to declared adjusted families; secondary 95% intervals have no family-wide claim. User metrics are averaged over five seeds before resampling.

S3 Controlled extension E1: complete utility

Table S6: All retained E1 utility rows.

Model	Seeds	NDCG@10	HR@10	MRR@10
FixedB r16 nodp	5	0.05516 ± 0.00495	0.10722 ± 0.00862	0.03958 ± 0.00408
FixedB r16 clip	5	0.05462 ± 0.00367	0.10679 ± 0.00845	0.03896 ± 0.00243
FixedB r16 $\epsilon = 8$	5	0.04492 ± 0.00830	0.08450 ± 0.02081	0.03296 ± 0.00499
FixedB r16 $\epsilon = 4$	5	0.03744 ± 0.00965	0.07006 ± 0.02161	0.02756 ± 0.00636
FixedB r16 $\epsilon = 2$	5	0.02529 ± 0.01052	0.04841 ± 0.02200	0.01834 ± 0.00729
FixedB r16 $\epsilon = 1$	5	0.01196 ± 0.00421	0.02463 ± 0.00860	0.00817 ± 0.00312
FixedB r32 nodp	5	0.05643 ± 0.00276	0.10955 ± 0.00378	0.04045 ± 0.00263
FixedB r32 clip	5	0.05581 ± 0.00312	0.10977 ± 0.00608	0.03962 ± 0.00241
FixedB r32 $\epsilon = 8$	5	0.04898 ± 0.00362	0.09342 ± 0.00856	0.03563 ± 0.00263
FixedB r32 $\epsilon = 4$	5	0.03896 ± 0.00534	0.07325 ± 0.01180	0.02865 ± 0.00397
FixedB r32 $\epsilon = 2$	5	0.02156 ± 0.00536	0.03970 ± 0.01130	0.01604 ± 0.00383
FixedB r32 $\epsilon = 1$	5	0.00839 ± 0.00241	0.01783 ± 0.00559	0.00558 ± 0.00169
FixedB r4 nodp	5	0.05211 ± 0.00161	0.09915 ± 0.00726	0.03796 ± 0.00175
FixedB r4 clip	5	0.05043 ± 0.00228	0.09618 ± 0.00161	0.03675 ± 0.00252
FixedB r4 $\epsilon = 8$	5	0.04531 ± 0.00484	0.08280 ± 0.00833	0.03398 ± 0.00468
FixedB r4 $\epsilon = 4$	5	0.03876 ± 0.00696	0.07197 ± 0.01008	0.02866 ± 0.00606
FixedB r4 $\epsilon = 2$	5	0.02480 ± 0.00733	0.04395 ± 0.01381	0.01894 ± 0.00545
FixedB r4 $\epsilon = 1$	5	0.01053 ± 0.00363	0.01975 ± 0.00644	0.00779 ± 0.00297
FixedB r8 nodp	5	0.05597 ± 0.00228	0.10679 ± 0.00422	0.04066 ± 0.00193
FixedB r8 clip	5	0.05485 ± 0.00177	0.10425 ± 0.00564	0.03999 ± 0.00104
FixedB r8 $\epsilon = 8$	5	0.04310 ± 0.00917	0.08217 ± 0.01824	0.03130 ± 0.00655
FixedB r8 $\epsilon = 4$	5	0.03564 ± 0.01190	0.06709 ± 0.02321	0.02617 ± 0.00871
FixedB r8 $\epsilon = 2$	5	0.02218 ± 0.01160	0.04013 ± 0.01850	0.01675 ± 0.00942
FixedB r8 $\epsilon = 1$	5	0.00904 ± 0.00536	0.01911 ± 0.00846	0.00604 ± 0.00442
Full nodp	5	0.05789 ± 0.00163	0.10934 ± 0.00360	0.04236 ± 0.00168
Full clip	5	0.05885 ± 0.00230	0.11529 ± 0.00529	0.04196 ± 0.00209
Full $\epsilon = 8$	5	0.05227 ± 0.00321	0.09894 ± 0.00630	0.03810 ± 0.00237
Full $\epsilon = 4$	5	0.04217 ± 0.00491	0.07856 ± 0.00819	0.03110 ± 0.00442
Full $\epsilon = 2$	5	0.02543 ± 0.00316	0.04565 ± 0.00859	0.01934 ± 0.00186
Full $\epsilon = 1$	5	0.01146 ± 0.00095	0.02166 ± 0.00178	0.00838 ± 0.00109
Two r16 nodp	5	0.05063 ± 0.00328	0.09766 ± 0.00601	0.03649 ± 0.00282
Two r16 clip	5	0.04522 ± 0.00777	0.08747 ± 0.01559	0.03250 ± 0.00554
Two r16 $\epsilon = 8$	5	0.04405 ± 0.00435	0.08344 ± 0.00851	0.03225 ± 0.00341
Two r16 $\epsilon = 4$	5	0.04420 ± 0.00583	0.08514 ± 0.01112	0.03185 ± 0.00434
Two r16 $\epsilon = 2$	5	0.03322 ± 0.00913	0.06306 ± 0.02233	0.02417 ± 0.00527
Two r16 $\epsilon = 1$	5	0.01496 ± 0.00303	0.02760 ± 0.00646	0.01116 ± 0.00212
Two r32 nodp	5	0.04861 ± 0.00206	0.09342 ± 0.00336	0.03510 ± 0.00207
Two r32 clip	5	0.05179 ± 0.00200	0.10382 ± 0.00528	0.03616 ± 0.00129
Two r32 $\epsilon = 8$	5	0.05116 ± 0.00508	0.09979 ± 0.01013	0.03657 ± 0.00369
Two r32 $\epsilon = 4$	5	0.04581 ± 0.00173	0.08981 ± 0.00256	0.03258 ± 0.00232
Two r32 $\epsilon = 2$	5	0.02897 ± 0.00489	0.05839 ± 0.00849	0.02015 ± 0.00413
Two r32 $\epsilon = 1$	5	0.00743 ± 0.00305	0.01571 ± 0.00506	0.00499 ± 0.00251
Two r4 nodp	5	0.04209 ± 0.01590	0.08301 ± 0.03212	0.02978 ± 0.01127
Two r4 clip	5	0.04561 ± 0.00358	0.08896 ± 0.00378	0.03264 ± 0.00351
Two r4 $\epsilon = 8$	5	0.04385 ± 0.00693	0.08408 ± 0.01385	0.03181 ± 0.00521
Two r4 $\epsilon = 4$	5	0.04341 ± 0.00452	0.08259 ± 0.00681	0.03162 ± 0.00383

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Table S6 (continued)

Model	Seeds	NDCG@10	HR@10	MRR@10
Two r4 $\epsilon = 2$	5	0.03332 ± 0.01056	0.06030 ± 0.02067	0.02515 ± 0.00755
Two r4 $\epsilon = 1$	5	0.02514 ± 0.00948	0.04777 ± 0.01731	0.01831 ± 0.00733
Two r8 nodp	5	0.05092 ± 0.00391	0.09830 ± 0.00838	0.03667 ± 0.00278
Two r8 clip	5	0.04851 ± 0.00477	0.09214 ± 0.00844	0.03539 ± 0.00399
Two r8 $\epsilon = 8$	5	0.04800 ± 0.00334	0.09214 ± 0.00771	0.03472 ± 0.00287
Two r8 $\epsilon = 4$	5	0.04423 ± 0.00408	0.08365 ± 0.01089	0.03227 ± 0.00216
Two r8 $\epsilon = 2$	5	0.04084 ± 0.00638	0.07749 ± 0.01370	0.02965 ± 0.00449
Two r8 $\epsilon = 1$	5	0.02096 ± 0.00827	0.03885 ± 0.01464	0.01553 ± 0.00638
Pop bounded nodp	1	0.04988 ± 0.00000	0.09766 ± 0.00000	0.03558 ± 0.00000
Pop $\epsilon = 8$	5	0.04955 ± 0.00054	0.09639 ± 0.00174	0.03550 ± 0.00028
Pop $\epsilon = 4$	5	0.04943 ± 0.00089	0.09575 ± 0.00275	0.03553 ± 0.00054
Pop $\epsilon = 2$	5	0.04951 ± 0.00064	0.09703 ± 0.00220	0.03529 ± 0.00042
Pop $\epsilon = 1$	5	0.04839 ± 0.00112	0.09597 ± 0.00266	0.03411 ± 0.00101

Entries are mean ± seed SD. The bounded no-noise popularity reference is deterministic; its SD is not training variability.

S3.1 Primary and secondary E1 comparisons

The primary family is `primary_fixed_vs_two`. It contains zero positive adjusted intervals, eight negative intervals, and eight including zero. The many secondary rows document clipping, noise, popularity, historical comparisons, and selected ranks; they must not be pooled into a larger set of confirmatory discoveries.

Table S7: E1 paired-user contrasts, including secondary families.

Contrast ($a - b$)	Difference	95% interval	Adjusted interval	Decision
primary_fixed_vs_two				
FixedB r4 $\epsilon = 8$ − Two r4 $\epsilon = 8$	0.001457	[-0.002380, 0.005282]	[-0.004377, 0.007183]	Includes zero
FixedB r4 $\epsilon = 4$ − Two r4 $\epsilon = 4$	-0.004653	[-0.009744, 0.000261]	[-0.012448, 0.002707]	Includes zero
FixedB r4 $\epsilon = 2$ − Two r4 $\epsilon = 2$	-0.008520	[-0.013855, -0.003479]	[-0.016681, -0.001036]	Negative
FixedB r4 $\epsilon = 1$ − Two r4 $\epsilon = 1$	-0.014601	[-0.019366, -0.010077]	[-0.021921, -0.007943]	Negative
FixedB r8 $\epsilon = 8$ − Two r8 $\epsilon = 8$	-0.004900	[-0.008914, -0.000958]	[-0.010988, 0.001048]	Includes zero
FixedB r8 $\epsilon = 4$ − Two r8 $\epsilon = 4$	-0.008589	[-0.013343, -0.003937]	[-0.015866, -0.001579]	Negative
FixedB r8 $\epsilon = 2$ − Two r8 $\epsilon = 2$	-0.018662	[-0.024407, -0.013119]	[-0.027695, -0.010368]	Negative
FixedB r8 $\epsilon = 1$ − Two r8 $\epsilon = 1$	-0.011923	[-0.016191, -0.007846]	[-0.018388, -0.005945]	Negative
FixedB r16 $\epsilon = 8$ − Two r16 $\epsilon = 8$	0.000876	[-0.002781, 0.004540]	[-0.004656, 0.006413]	Includes zero
FixedB r16 $\epsilon = 4$ − Two r16 $\epsilon = 4$	-0.006756	[-0.011228, -0.002414]	[-0.013635, -0.000322]	Negative
FixedB r16 $\epsilon = 2$ − Two r16 $\epsilon = 2$	-0.007933	[-0.012971, -0.003079]	[-0.015604, -0.000732]	Negative
FixedB r16 $\epsilon = 1$ − Two r16 $\epsilon = 1$	-0.003002	[-0.006808, 0.000617]	[-0.008837, 0.002435]	Includes zero
FixedB r32 $\epsilon = 8$ − Two r32 $\epsilon = 8$	-0.002180	[-0.006771, 0.002314]	[-0.009253, 0.004501]	Includes zero
FixedB r32 $\epsilon = 4$ − Two r32 $\epsilon = 4$	-0.006848	[-0.011976, -0.001893]	[-0.014640, 0.000575]	Includes zero
FixedB r32 $\epsilon = 2$ − Two r32 $\epsilon = 2$	-0.007408	[-0.011629, -0.003215]	[-0.013724, -0.001159]	Negative
FixedB r32 $\epsilon = 1$ − Two r32 $\epsilon = 1$	0.000958	[-0.001755, 0.003659]	[-0.003118, 0.005012]	Includes zero
secondary_fixed_vs_full				
FixedB r4 $\epsilon = 8$ − Full $\epsilon = 8$	-0.006959	[-0.011403, -0.002593]	—	Secondary
FixedB r4 $\epsilon = 4$ − Full $\epsilon = 4$	-0.003409	[-0.008298, 0.001435]	—	Secondary
FixedB r4 $\epsilon = 2$ − Full $\epsilon = 2$	-0.000631	[-0.005531, 0.004278]	—	Secondary
FixedB r4 $\epsilon = 1$ − Full $\epsilon = 1$	-0.000926	[-0.004631, 0.002803]	—	Secondary
FixedB r8 $\epsilon = 8$ − Full $\epsilon = 8$	-0.009175	[-0.013749, -0.004718]	—	Secondary
FixedB r8 $\epsilon = 4$ − Full $\epsilon = 4$	-0.006530	[-0.011384, -0.001692]	—	Secondary
FixedB r8 $\epsilon = 2$ − Full $\epsilon = 2$	-0.003249	[-0.007716, 0.001211]	—	Secondary
FixedB r8 $\epsilon = 1$ − Full $\epsilon = 1$	-0.002419	[-0.005703, 0.000801]	—	Secondary
FixedB r16 $\epsilon = 8$ − Full $\epsilon = 8$	-0.007348	[-0.011625, -0.003129]	—	Secondary
FixedB r16 $\epsilon = 4$ − Full $\epsilon = 4$	-0.004726	[-0.009132, -0.000334]	—	Secondary

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Table S7 (continued)

Contrast ($a - b$)	Difference	95% interval	Adjusted interval	Decision
FixedB r16 $\epsilon = 2$ − Full $\epsilon = 2$	-0.000142	[-0.004546, 0.004209]	—	Secondary
FixedB r16 $\epsilon = 1$ − Full $\epsilon = 1$	0.000497	[-0.002939, 0.003991]	—	Secondary
FixedB r32 $\epsilon = 8$ − Full $\epsilon = 8$	-0.003293	[-0.007110, 0.000502]	—	Secondary
FixedB r32 $\epsilon = 4$ − Full $\epsilon = 4$	-0.003210	[-0.007452, 0.000993]	—	Secondary
FixedB r32 $\epsilon = 2$ − Full $\epsilon = 2$	-0.003869	[-0.008006, 0.000253]	—	Secondary
FixedB r32 $\epsilon = 1$ − Full $\epsilon = 1$	-0.003076	[-0.006307, 0.000126]	—	Secondary
secondary_fixed_vs_pop				
FixedB r4 $\epsilon = 8$ − Pop $\epsilon = 8$	-0.004236	[-0.010563, 0.001935]	—	Secondary
FixedB r4 $\epsilon = 4$ − Pop $\epsilon = 4$	-0.010668	[-0.017792, -0.003837]	—	Secondary
FixedB r4 $\epsilon = 2$ − Pop $\epsilon = 2$	-0.024709	[-0.033161, -0.016674]	—	Secondary
FixedB r4 $\epsilon = 1$ − Pop $\epsilon = 1$	-0.037855	[-0.047721, -0.028521]	—	Secondary
FixedB r8 $\epsilon = 8$ − Pop $\epsilon = 8$	-0.006451	[-0.013278, 0.000253]	—	Secondary
FixedB r8 $\epsilon = 4$ − Pop $\epsilon = 4$	-0.013788	[-0.021102, -0.006722]	—	Secondary
FixedB r8 $\epsilon = 2$ − Pop $\epsilon = 2$	-0.027328	[-0.035865, -0.019213]	—	Secondary
FixedB r8 $\epsilon = 1$ − Pop $\epsilon = 1$	-0.039347	[-0.049148, -0.030065]	—	Secondary
FixedB r16 $\epsilon = 8$ − Pop $\epsilon = 8$	-0.004625	[-0.010626, 0.001175]	—	Secondary
FixedB r16 $\epsilon = 4$ − Pop $\epsilon = 4$	-0.011985	[-0.018717, -0.005541]	—	Secondary
FixedB r16 $\epsilon = 2$ − Pop $\epsilon = 2$	-0.024221	[-0.032614, -0.016318]	—	Secondary
FixedB r16 $\epsilon = 1$ − Pop $\epsilon = 1$	-0.036431	[-0.046080, -0.027364]	—	Secondary
FixedB r32 $\epsilon = 8$ − Pop $\epsilon = 8$	-0.000570	[-0.006490, 0.005259]	—	Secondary
FixedB r32 $\epsilon = 4$ − Pop $\epsilon = 4$	-0.010468	[-0.017463, -0.003739]	—	Secondary
FixedB r32 $\epsilon = 2$ − Pop $\epsilon = 2$	-0.027948	[-0.036692, -0.019677]	—	Secondary
FixedB r32 $\epsilon = 1$ − Pop $\epsilon = 1$	-0.040004	[-0.050126, -0.030477]	—	Secondary
secondary_fixed_vs_historical_B4				
FixedB r4 $\epsilon = 8$ − Hist. B4 r4 $\epsilon = 8$	0.006767	[0.002552, 0.011073]	—	Secondary
FixedB r4 $\epsilon = 4$ − Hist. B4 r4 $\epsilon = 4$	0.000914	[-0.003172, 0.004880]	—	Secondary
FixedB r4 $\epsilon = 2$ − Hist. B4 r4 $\epsilon = 2$	-0.004281	[-0.008674, 0.000019]	—	Secondary
FixedB r4 $\epsilon = 1$ − Hist. B4 r4 $\epsilon = 1$	-0.005442	[-0.009071, -0.001841]	—	Secondary
FixedB r8 $\epsilon = 8$ − Hist. B4 r8 $\epsilon = 8$	-0.002289	[-0.006428, 0.001803]	—	Secondary
FixedB r8 $\epsilon = 4$ − Hist. B4 r8 $\epsilon = 4$	-0.007013	[-0.011821, -0.002273]	—	Secondary
FixedB r8 $\epsilon = 2$ − Hist. B4 r8 $\epsilon = 2$	-0.012306	[-0.017375, -0.007384]	—	Secondary
FixedB r8 $\epsilon = 1$ − Hist. B4 r8 $\epsilon = 1$	-0.009120	[-0.013339, -0.005114]	—	Secondary
FixedB r16 $\epsilon = 8$ − Hist. B4 r16 $\epsilon = 8$	0.001295	[-0.002859, 0.005485]	—	Secondary
FixedB r16 $\epsilon = 4$ − Hist. B4 r16 $\epsilon = 4$	-0.000831	[-0.004957, 0.003290]	—	Secondary
FixedB r16 $\epsilon = 2$ − Hist. B4 r16 $\epsilon = 2$	-0.008707	[-0.013595, -0.003995]	—	Secondary
FixedB r16 $\epsilon = 1$ − Hist. B4 r16 $\epsilon = 1$	-0.004726	[-0.008642, -0.000980]	—	Secondary
FixedB r32 $\epsilon = 8$ − Hist. B4 r32 $\epsilon = 8$	0.004956	[0.000818, 0.009115]	—	Secondary

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Table S7 (continued)

Contrast ($a - b$)	Difference	95% interval	Adjusted interval	Decision
FixedB r32 $\epsilon = 4$ − Hist. B4 r32 $\epsilon = 4$	-0.005529	[-0.010551, -0.000618]	—	Secondary
FixedB r32 $\epsilon = 2$ − Hist. B4 r32 $\epsilon = 2$	-0.014945	[-0.020199, -0.009914]	—	Secondary
FixedB r32 $\epsilon = 1$ − Hist. B4 r32 $\epsilon = 1$	-0.007005	[-0.010557, -0.003611]	—	Secondary
secondary_clipping				
Full clip − Full nodp	0.000957	[-0.001498, 0.003390]	—	Secondary
Two r4 clip − Two r4 nodp	0.003519	[0.000065, 0.007042]	—	Secondary
Two r8 clip − Two r8 nodp	-0.002406	[-0.005021, 0.000125]	—	Secondary
Two r16 clip − Two r16 nodp	-0.005417	[-0.008998, -0.001955]	—	Secondary
Two r32 clip − Two r32 nodp	0.003179	[0.000296, 0.006130]	—	Secondary
FixedB r4 clip − FixedB r4 nodp	-0.001678	[-0.003689, 0.000304]	—	Secondary
FixedB r8 clip − FixedB r8 nodp	-0.001123	[-0.003019, 0.000753]	—	Secondary
FixedB r16 clip − FixedB r16 nodp	-0.000541	[-0.002598, 0.001495]	—	Secondary
FixedB r32 clip − FixedB r32 nodp	-0.000625	[-0.002791, 0.001545]	—	Secondary
secondary_noise				
Full $\epsilon = 8$ − Full clip	-0.006575	[-0.011293, -0.002020]	—	Secondary
Full $\epsilon = 4$ − Full clip	-0.016676	[-0.023324, -0.010332]	—	Secondary
Full $\epsilon = 2$ − Full clip	-0.033417	[-0.042650, -0.024648]	—	Secondary
Full $\epsilon = 1$ − Full clip	-0.047385	[-0.058365, -0.036894]	—	Secondary
Two r4 $\epsilon = 8$ − Two r4 clip	-0.001751	[-0.004555, 0.000980]	—	Secondary
Two r4 $\epsilon = 4$ − Two r4 clip	-0.002192	[-0.005371, 0.000871]	—	Secondary
Two r4 $\epsilon = 2$ − Two r4 clip	-0.012287	[-0.016729, -0.008085]	—	Secondary
Two r4 $\epsilon = 1$ − Two r4 clip	-0.020469	[-0.026293, -0.014895]	—	Secondary
Two r8 $\epsilon = 8$ − Two r8 clip	-0.000516	[-0.003294, 0.002288]	—	Secondary
Two r8 $\epsilon = 4$ − Two r8 clip	-0.004283	[-0.008233, -0.000387]	—	Secondary
Two r8 $\epsilon = 2$ − Two r8 clip	-0.007671	[-0.012226, -0.003217]	—	Secondary
Two r8 $\epsilon = 1$ − Two r8 clip	-0.027547	[-0.034818, -0.020596]	—	Secondary
Two r16 $\epsilon = 8$ − Two r16 clip	-0.001170	[-0.004517, 0.002197]	—	Secondary
Two r16 $\epsilon = 4$ − Two r16 clip	-0.001018	[-0.005047, 0.003000]	—	Secondary
Two r16 $\epsilon = 2$ − Two r16 clip	-0.011998	[-0.017180, -0.007029]	—	Secondary
Two r16 $\epsilon = 1$ − Two r16 clip	-0.030257	[-0.037731, -0.023041]	—	Secondary
Two r32 $\epsilon = 8$ − Two r32 clip	-0.000635	[-0.004645, 0.003312]	—	Secondary
Two r32 $\epsilon = 4$ − Two r32 clip	-0.005985	[-0.011136, -0.000917]	—	Secondary
Two r32 $\epsilon = 2$ − Two r32 clip	-0.022826	[-0.031178, -0.014792]	—	Secondary
Two r32 $\epsilon = 1$ − Two r32 clip	-0.044365	[-0.054693, -0.034404]	—	Secondary
FixedB r4 $\epsilon = 8$ − FixedB r4 clip	-0.005122	[-0.008528, -0.001876]	—	Secondary
FixedB r4 $\epsilon = 4$ − FixedB r4 clip	-0.011674	[-0.016478, -0.007189]	—	Secondary
FixedB r4 $\epsilon = 2$ − FixedB r4 clip	-0.025636	[-0.032368, -0.019284]	—	Secondary
FixedB r4 $\epsilon = 1$ − FixedB r4 clip	-0.039899	[-0.048654, -0.031598]	—	Secondary

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Table S7 (continued)

Contrast ($a - b$)	Difference	95% interval	Adjusted interval	Decision
FixedB r8 $\epsilon = 8$ − FixedB r8 clip	-0.011753	[-0.016065, -0.007565]	—	Secondary
FixedB r8 $\epsilon = 4$ − FixedB r8 clip	-0.019209	[-0.024866, -0.013732]	—	Secondary
FixedB r8 $\epsilon = 2$ − FixedB r8 clip	-0.032670	[-0.040481, -0.025158]	—	Secondary
FixedB r8 $\epsilon = 1$ − FixedB r8 clip	-0.045807	[-0.055368, -0.036663]	—	Secondary
FixedB r16 $\epsilon = 8$ − FixedB r16 clip	-0.009695	[-0.013966, -0.005556]	—	Secondary
FixedB r16 $\epsilon = 4$ − FixedB r16 clip	-0.017175	[-0.022614, -0.011982]	—	Secondary
FixedB r16 $\epsilon = 2$ − FixedB r16 clip	-0.029332	[-0.036816, -0.022200]	—	Secondary
FixedB r16 $\epsilon = 1$ − FixedB r16 clip	-0.042660	[-0.052120, -0.033609]	—	Secondary
FixedB r32 $\epsilon = 8$ − FixedB r32 clip	-0.006829	[-0.010844, -0.002952]	—	Secondary
FixedB r32 $\epsilon = 4$ − FixedB r32 clip	-0.016847	[-0.022699, -0.011260]	—	Secondary
FixedB r32 $\epsilon = 2$ − FixedB r32 clip	-0.034248	[-0.042474, -0.026333]	—	Secondary
FixedB r32 $\epsilon = 1$ − FixedB r32 clip	-0.047422	[-0.057748, -0.037519]	—	Secondary
secondary_pop_noise				
Pop $\epsilon = 8$ − Pop bounded nodp	-0.000337	[-0.000954, 0.000219]	—	Secondary
Pop $\epsilon = 4$ − Pop bounded nodp	-0.000457	[-0.001374, 0.000467]	—	Secondary
Pop $\epsilon = 2$ − Pop bounded nodp	-0.000378	[-0.001815, 0.001035]	—	Secondary
Pop $\epsilon = 1$ − Pop bounded nodp	-0.001495	[-0.003619, 0.000529]	—	Secondary
secondary_selected_ranks				
FixedB r32 $\epsilon = 8$ − Two r32 $\epsilon = 8$	-0.002180	[-0.006771, 0.002314]	—	Secondary
FixedB r32 $\epsilon = 4$ − Two r32 $\epsilon = 4$	-0.006848	[-0.011976, -0.001893]	—	Secondary
FixedB r16 $\epsilon = 2$ − Two r8 $\epsilon = 2$	-0.015554	[-0.020994, -0.010291]	—	Secondary
FixedB r16 $\epsilon = 1$ − Two r4 $\epsilon = 1$	-0.013178	[-0.017875, -0.008707]	—	Secondary

Decisions refer only to declared adjusted families; secondary 95% intervals have no family-wide claim. User metrics are averaged over five seeds before resampling.

S4 E1 geometry and local personalization

S4.1 Effective matrix and score diagnostics

Coordinate dimension, Frobenius noise norm, entrywise score RMSE, and clipping fraction have different units and meanings. The matrix-noise term is conditioned on the signal-updated factors; local scale can amplify score disturbance. Large finite values remain included. The expected bilinear fraction is not a causal percentage of NDCG loss.

Table S8: E1 effective-noise and clipping diagnostics across all 54 settings.

Unit	Level	Q noise norm	Score RMSE	Cross (%)	Clipped (%)	Mean $\|p\|$
FixedB r16	No DP	0.0000	0	0.0000	0.00	1.601×10^{0}
FixedB r16	Clip only	0.0000	0	0.0000	18.56	1.446×10^{0}
FixedB r16	8	1.4023	6.646×10^{-3}	0.0000	23.73	1.717×10^{0}
FixedB r16	4	2.0065	1.043×10^{-2}	0.0000	27.21	1.894×10^{0}
FixedB r16	2	3.0664	1.817×10^{-2}	0.0000	32.53	2.145×10^{0}
FixedB r16	1	5.1863	3.594×10^{-2}	0.0000	40.32	2.404×10^{0}
FixedB r32	No DP	0.0000	0	0.0000	0.00	1.643×10^{0}
FixedB r32	Clip only	0.0000	0	0.0000	19.42	1.483×10^{0}
FixedB r32	8	1.9828	7.380×10^{-3}	0.0000	27.37	1.905×10^{0}
FixedB r32	4	2.8370	1.186×10^{-2}	0.0000	31.81	2.137×10^{0}
FixedB r32	2	4.3357	2.112×10^{-2}	0.0000	38.23	2.454×10^{0}
FixedB r32	1	7.3331	4.271×10^{-2}	0.0000	46.85	2.796×10^{0}
FixedB r4	No DP	0.0000	0	0.0000	0.00	1.448×10^{0}
FixedB r4	Clip only	0.0000	0	0.0000	16.81	1.305×10^{0}
FixedB r4	8	0.7008	5.532×10^{-3}	0.0000	18.20	1.354×10^{0}
FixedB r4	4	1.0028	8.200×10^{-3}	0.0000	19.35	1.402×10^{0}
FixedB r4	2	1.5325	1.331×10^{-2}	0.0000	21.67	1.486×10^{0}
FixedB r4	1	2.5920	2.450×10^{-2}	0.0000	25.58	1.570×10^{0}
FixedB r8	No DP	0.0000	0	0.0000	0.00	1.611×10^{0}
FixedB r8	Clip only	0.0000	0	0.0000	18.89	1.455×10^{0}
FixedB r8	8	0.9916	6.078×10^{-3}	0.0000	20.77	1.529×10^{0}
FixedB r8	4	1.4188	9.133×10^{-3}	0.0000	22.56	1.607×10^{0}
FixedB r8	2	2.1683	1.522×10^{-2}	0.0000	26.18	1.745×10^{0}
FixedB r8	1	3.6673	2.912×10^{-2}	0.0000	31.98	1.905×10^{0}
Full	No DP	0.0000	0	0.0000	0.00	1.611×10^{0}
Full	Clip only	0.0000	0	0.0000	19.31	1.452×10^{0}
Full	8	2.8035	8.507×10^{-3}	0.0000	32.32	2.195×10^{0}
Full	4	4.0114	1.393×10^{-2}	0.0000	37.67	2.503×10^{0}
Full	2	6.1305	2.511×10^{-2}	0.0000	45.23	2.894×10^{0}
Full	1	10.3686	5.110×10^{-2}	0.0000	54.12	3.310×10^{0}
Two r16	No DP	0.0000	0	0.0000	0.00	1.391×10^{0}
Two r16	Clip only	0.0000	0	0.0000	16.26	1.121×10^{0}
Two r16	8	0.8087	4.374×10^{-3}	0.0878	23.94	1.468×10^{0}
Two r16	4	1.3376	8.538×10^{-3}	0.1346	31.30	1.772×10^{0}
Two r16	2	2.5988	1.828×10^{-2}	0.1945	41.74	2.115×10^{0}
Two r16	1	6.5423	5.266×10^{-2}	0.2511	59.30	2.488×10^{0}
Two r32	No DP	0.0000	0	0.0000	0.00	2.029×10^{0}
Two r32	Clip only	0.0000	0	0.0000	26.11	1.785×10^{0}
Two r32	8	3.0434	2.433×10^{-2}	0.1984	47.41	2.469×10^{0}
Two r32	4	5.6858	4.308×10^{-2}	0.2382	58.60	2.706×10^{0}

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Table S8 (continued)

Unit	Level	Q noise norm	Score RMSE	Cross (%)	Clipped (%)	Mean $\|p\|$
Two r32	2	12.3862	1.173×10^{-1}	0.2738	69.74	2.978×10^{0}
Two r32	1	33.7884	9.846×10^{16}	0.3010	82.51	6.179×10^{16}
Two r4	No DP	0.0000	0	0.0000	0.00	2.961×10^{5}
Two r4	Clip only	0.0000	0	0.0000	25.85	1.525×10^{0}
Two r4	8	0.5531	1.028×10^{-2}	0.0469	30.93	1.773×10^{0}
Two r4	4	0.8674	1.488×10^{-2}	0.0800	35.68	1.925×10^{0}
Two r4	2	1.5426	2.622×10^{-2}	0.1380	40.72	1.981×10^{0}
Two r4	1	3.6017	2.248×10^{0}	0.2071	54.61	6.070×10^{0}
Two r8	No DP	0.0000	0	0.0000	0.00	1.756×10^{0}
Two r8	Clip only	0.0000	0	0.0000	22.18	1.375×10^{0}
Two r8	8	0.6660	5.646×10^{-3}	0.0648	28.32	1.643×10^{0}
Two r8	4	1.0609	9.210×10^{-3}	0.1070	32.37	1.784×10^{0}
Two r8	2	1.9937	1.961×10^{-2}	0.1654	41.81	2.072×10^{0}
Two r8	1	4.8173	5.554×10^{-2}	0.2317	56.45	2.315×10^{0}

Q noise norm is the mean over seeds of per-run root-mean-squared Frobenius norms, not entrywise RMS. Cross is the expected bilinear fraction of matrix-noise energy. Score RMSE can be dominated by large finite local vectors.

Table S9: E1 synthetic checks of conditional moment calculations.

Method	τ	Draws	Q: measured/expected	Scores: measured/expected
Two	0.005	2000	1.001147	1.000056
FixedB	0.005	2000	0.998374	0.999316
Two	0.015	2000	0.999972	0.997460
FixedB	0.015	2000	0.999621	1.000353
Two	0.030	2000	1.002638	1.001816
FixedB	0.030	2000	0.999637	0.997283
Two	0.100	2000	1.001573	1.002951
FixedB	0.100	2000	1.002546	1.000933

Eight synthetic cases, 2,000 draws each. Finite Monte Carlo discrepancies are expected; these are implementation checks rather than real-data utility results.

S4.2 Activity and cold-target results

Activity thresholds are at most 22, 23–61, and at least 62 training positives. The test set has only fourteen cold targets. These tables are descriptive, with no adjusted subgroup superiority claim. Cold items remain in the main evaluation even when no method retrieves them in the top ten.

Table S10: Descriptive activity and cold-target NDCG@10, including historical comparators.

Model	Low (319)	Medium (310)	High (313)	Cold (14)	Warm (928)
FixedB r16 nodp	0.07789	0.03667	0.05030	0.00000	0.05599
FixedB r16 clip	0.07860	0.03705	0.04757	0.00000	0.05544
FixedB r16 $\epsilon = 8$	0.06380	0.03290	0.03759	0.00000	0.04560
FixedB r16 $\epsilon = 4$	0.04842	0.02915	0.03447	0.00000	0.03801
FixedB r16 $\epsilon = 2$	0.02877	0.02017	0.02680	0.00000	0.02567
FixedB r16 $\epsilon = 1$	0.01107	0.01030	0.01451	0.00000	0.01214
FixedB r32 nodp	0.08077	0.03802	0.04986	0.00000	0.05728
FixedB r32 clip	0.08055	0.03776	0.04847	0.00000	0.05665
FixedB r32 $\epsilon = 8$	0.06297	0.03588	0.04769	0.00000	0.04972
FixedB r32 $\epsilon = 4$	0.04813	0.03055	0.03794	0.00000	0.03955
FixedB r32 $\epsilon = 2$	0.02511	0.01976	0.01972	0.00000	0.02188
FixedB r32 $\epsilon = 1$	0.00967	0.01053	0.00496	0.00000	0.00851
FixedB r4 nodp	0.06272	0.03690	0.05637	0.00000	0.05290
FixedB r4 clip	0.05995	0.03770	0.05335	0.00000	0.05119
FixedB r4 $\epsilon = 8$	0.05563	0.03461	0.04540	0.00000	0.04600
FixedB r4 $\epsilon = 4$	0.04953	0.03000	0.03645	0.00000	0.03934
FixedB r4 $\epsilon = 2$	0.03008	0.02039	0.02378	0.00000	0.02517
FixedB r4 $\epsilon = 1$	0.01250	0.00892	0.01013	0.00000	0.01069
FixedB r8 nodp	0.07976	0.03555	0.05195	0.00000	0.05682
FixedB r8 clip	0.07835	0.03643	0.04913	0.00000	0.05568
FixedB r8 $\epsilon = 8$	0.06310	0.02840	0.03727	0.00000	0.04375
FixedB r8 $\epsilon = 4$	0.05349	0.02257	0.03039	0.00000	0.03618
FixedB r8 $\epsilon = 2$	0.03167	0.01633	0.01830	0.00000	0.02251
FixedB r8 $\epsilon = 1$	0.01199	0.00541	0.00964	0.00430	0.00911
Full nodp	0.07961	0.03828	0.05517	0.00000	0.05876
Full clip	0.08164	0.04184	0.05245	0.00000	0.05973
Full $\epsilon = 8$	0.07144	0.03715	0.04770	0.00000	0.05306
Full $\epsilon = 4$	0.05504	0.03223	0.03890	0.00000	0.04281
Full $\epsilon = 2$	0.02906	0.02108	0.02603	0.00430	0.02575
Full $\epsilon = 1$	0.01338	0.01024	0.01071	0.00000	0.01163
Two r16 nodp	0.06565	0.03342	0.05238	0.00000	0.05140
Two r16 clip	0.06157	0.03352	0.04014	0.00000	0.04590
Two r16 $\epsilon = 8$	0.05878	0.03342	0.03955	0.00000	0.04471
Two r16 $\epsilon = 4$	0.05464	0.03103	0.04659	0.00000	0.04487
Two r16 $\epsilon = 2$	0.03896	0.02526	0.03526	0.00451	0.03365
Two r16 $\epsilon = 1$	0.01278	0.01377	0.01836	0.00000	0.01519
Two r32 nodp	0.05918	0.03596	0.05037	0.00000	0.04935
Two r32 clip	0.06445	0.03705	0.05349	0.00000	0.05257
Two r32 $\epsilon = 8$	0.06602	0.03612	0.05090	0.00000	0.05193
Two r32 $\epsilon = 4$	0.05421	0.03638	0.04658	0.00000	0.04650
Two r32 $\epsilon = 2$	0.03618	0.02167	0.02885	0.00000	0.02940
Two r32 $\epsilon = 1$	0.00435	0.00844	0.00956	0.00000	0.00754
Two r4 nodp	0.05147	0.02861	0.04587	0.00000	0.04272

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Table S10 (continued)

Model	Low (319)	Medium (310)	High (313)	Cold (14)	Warm (928)
Two r4 clip	0.05129	0.03506	0.05026	0.00000	0.04629
Two r4 $\epsilon = 8$	0.05115	0.03420	0.04599	0.00000	0.04452
Two r4 $\epsilon = 4$	0.05040	0.03389	0.04573	0.00000	0.04407
Two r4 $\epsilon = 2$	0.03720	0.02408	0.03852	0.00000	0.03382
Two r4 $\epsilon = 1$	0.02663	0.02232	0.02640	0.00000	0.02552
Two r8 nodp	0.06509	0.03582	0.05143	0.00000	0.05169
Two r8 clip	0.06636	0.03425	0.04445	0.00000	0.04924
Two r8 $\epsilon = 8$	0.06336	0.03297	0.04722	0.00000	0.04872
Two r8 $\epsilon = 4$	0.05561	0.03013	0.04659	0.00000	0.04490
Two r8 $\epsilon = 2$	0.04872	0.02703	0.04649	0.00000	0.04146
Two r8 $\epsilon = 1$	0.02423	0.01806	0.02052	0.00451	0.02121
Pop bounded nodp	0.05585	0.03848	0.05510	0.00000	0.05064
Pop $\epsilon = 8$	0.05586	0.03805	0.05450	0.00000	0.05030
Pop $\epsilon = 4$	0.05599	0.03765	0.05440	0.00000	0.05017
Pop $\epsilon = 2$	0.05575	0.03687	0.05566	0.00000	0.05025
Pop $\epsilon = 1$	0.05576	0.03556	0.05359	0.00000	0.04912
Hist. B4 r4 $\epsilon = 1$	0.02182	0.01181	0.01415	0.00906	0.01608
Hist. B4 r4 $\epsilon = 2$	0.03150	0.02583	0.02982	0.00000	0.02952
Hist. B4 r4 $\epsilon = 4$	0.05180	0.02575	0.03561	0.00000	0.03842
Hist. B4 r4 $\epsilon = 8$	0.04988	0.02836	0.03708	0.00000	0.03913
Hist. B4 r4 nodp	0.04885	0.02873	0.04465	0.00000	0.04145
Hist. B4 r8 $\epsilon = 1$	0.02286	0.01334	0.01815	0.00000	0.01844
Hist. B4 r8 $\epsilon = 2$	0.04292	0.02552	0.03476	0.00000	0.03501
Hist. B4 r8 $\epsilon = 4$	0.05325	0.03121	0.04318	0.00000	0.04330
Hist. B4 r8 $\epsilon = 8$	0.05937	0.03293	0.04347	0.00000	0.04607
Hist. B4 r8 nodp	0.05913	0.03504	0.04974	0.00000	0.04881
Hist. B4 r16 $\epsilon = 1$	0.01653	0.01090	0.02257	0.00000	0.01694
Hist. B4 r16 $\epsilon = 2$	0.03402	0.03165	0.03628	0.00000	0.03451
Hist. B4 r16 $\epsilon = 4$	0.04943	0.02817	0.03691	0.00000	0.03885
Hist. B4 r16 $\epsilon = 8$	0.05584	0.03154	0.04316	0.00000	0.04429
Hist. B4 r16 nodp	0.06670	0.03302	0.04722	0.00000	0.04988
Hist. B4 r32 $\epsilon = 1$	0.01381	0.01632	0.01609	0.01066	0.01546
Hist. B4 r32 $\epsilon = 2$	0.04362	0.02698	0.03868	0.00000	0.03705
Hist. B4 r32 $\epsilon = 4$	0.05695	0.03257	0.04359	0.00000	0.04516
Hist. B4 r32 $\epsilon = 8$	0.05760	0.03264	0.04146	0.00000	0.04469
Hist. B4 r32 nodp	0.06620	0.03673	0.04944	0.00000	0.05170

No subgroup significance claim. Group sizes are users/test targets. Historical B4 rows are retained comparator rows; all model identities remain explicit.

S4.3 Validation personalization ablations

"Own" retains each user's vector. "Permuted" assigns vectors to different users. "Mean" substitutes the mean user vector. These measurements access internal states and diagnose personalization; they are not privacy-accounted deployment alternatives. Small differences should not be treated as formal significance findings.

Table S11: E1 validation personalization ablations (five-seed mean and SD).

Unit	Level	Own user vector	Permuted vector	Mean vector
FixedB r16	No DP	0.05035 ± 0.00333	0.04267 ± 0.00450	0.04698 ± 0.00209
FixedB r16	Clip only	0.05237 ± 0.00357	0.04325 ± 0.00374	0.04786 ± 0.00325
FixedB r16	8	0.04270 ± 0.00537	0.03393 ± 0.00681	0.04494 ± 0.00240
FixedB r16	4	0.03842 ± 0.00714	0.02975 ± 0.00752	0.04239 ± 0.00375
FixedB r16	2	0.02742 ± 0.00657	0.02139 ± 0.00693	0.03722 ± 0.00595
FixedB r16	1	0.01198 ± 0.00301	0.00977 ± 0.00294	0.02420 ± 0.01003
FixedB r32	No DP	0.05340 ± 0.00223	0.04290 ± 0.00240	0.04708 ± 0.00223
FixedB r32	Clip only	0.05581 ± 0.00254	0.04237 ± 0.00104	0.04836 ± 0.00184
FixedB r32	8	0.04900 ± 0.00274	0.03723 ± 0.00115	0.04670 ± 0.00161
FixedB r32	4	0.03922 ± 0.00166	0.03041 ± 0.00276	0.04479 ± 0.00148
FixedB r32	2	0.02135 ± 0.00090	0.01996 ± 0.00299	0.03966 ± 0.00185
FixedB r32	1	0.00846 ± 0.00311	0.00864 ± 0.00151	0.02454 ± 0.00369
FixedB r4	No DP	0.04655 ± 0.00345	0.04075 ± 0.00227	0.04666 ± 0.00051
FixedB r4	Clip only	0.04536 ± 0.00225	0.04091 ± 0.00318	0.04747 ± 0.00090
FixedB r4	8	0.03948 ± 0.00352	0.03462 ± 0.00302	0.04476 ± 0.00334
FixedB r4	4	0.03421 ± 0.00533	0.02975 ± 0.00624	0.04231 ± 0.00398
FixedB r4	2	0.02330 ± 0.00469	0.02221 ± 0.00637	0.03466 ± 0.00489
FixedB r4	1	0.01111 ± 0.00260	0.01047 ± 0.00314	0.02007 ± 0.00404
FixedB r8	No DP	0.05012 ± 0.00229	0.04206 ± 0.00391	0.04531 ± 0.00206
FixedB r8	Clip only	0.05172 ± 0.00340	0.04140 ± 0.00534	0.04576 ± 0.00180
FixedB r8	8	0.04172 ± 0.01147	0.03425 ± 0.00847	0.04410 ± 0.00374
FixedB r8	4	0.03284 ± 0.01164	0.02848 ± 0.00873	0.03995 ± 0.00560
FixedB r8	2	0.01998 ± 0.00933	0.01794 ± 0.00640	0.03353 ± 0.00775
FixedB r8	1	0.00858 ± 0.00304	0.00848 ± 0.00117	0.01891 ± 0.00409
Full	No DP	0.05314 ± 0.00257	0.04273 ± 0.00271	0.04742 ± 0.00131
Full	Clip only	0.05609 ± 0.00249	0.04413 ± 0.00302	0.04846 ± 0.00180
Full	8	0.04661 ± 0.00267	0.03401 ± 0.00103	0.04603 ± 0.00156
Full	4	0.03687 ± 0.00505	0.02763 ± 0.00196	0.04418 ± 0.00283
Full	2	0.02054 ± 0.00563	0.01994 ± 0.00328	0.04081 ± 0.00245
Full	1	0.00888 ± 0.00336	0.01115 ± 0.00260	0.03049 ± 0.00385
Two r16	No DP	0.04276 ± 0.00510	0.04129 ± 0.00551	0.04585 ± 0.00199
Two r16	Clip only	0.04076 ± 0.00778	0.03830 ± 0.00712	0.04471 ± 0.00501
Two r16	8	0.04242 ± 0.00541	0.03731 ± 0.00602	0.04330 ± 0.00271
Two r16	4	0.04313 ± 0.00574	0.03693 ± 0.00308	0.04582 ± 0.00263
Two r16	2	0.03521 ± 0.00696	0.03088 ± 0.00531	0.03946 ± 0.00437
Two r16	1	0.02057 ± 0.00431	0.01845 ± 0.00436	0.03264 ± 0.00746
Two r32	No DP	0.04819 ± 0.00235	0.04589 ± 0.00134	0.04707 ± 0.00128
Two r32	Clip only	0.04969 ± 0.00231	0.04682 ± 0.00251	0.04808 ± 0.00170
Two r32	8	0.04921 ± 0.00470	0.04277 ± 0.00119	0.04530 ± 0.00093
Two r32	4	0.04589 ± 0.00311	0.03877 ± 0.00373	0.04515 ± 0.00299
Two r32	2	0.02777 ± 0.00286	0.02274 ± 0.00448	0.03647 ± 0.00222
Two r32	1	0.00870 ± 0.00186	0.00590 ± 0.00367	0.00740 ± 0.00722

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Table S11 (continued)

Unit	Level	Own user vector	Permuted vector	Mean vector
Two r4	No DP	0.03670 ± 0.01435	0.03487 ± 0.01394	0.03954 ± 0.01683
Two r4	Clip only	0.04235 ± 0.00127	0.03945 ± 0.00440	0.04708 ± 0.00114
Two r4	8	0.04170 ± 0.00317	0.03973 ± 0.00346	0.04466 ± 0.00279
Two r4	4	0.04151 ± 0.00359	0.03929 ± 0.00257	0.04367 ± 0.00213
Two r4	2	0.03201 ± 0.01201	0.02821 ± 0.01000	0.03741 ± 0.00684
Two r4	1	0.02154 ± 0.01115	0.01940 ± 0.01008	0.02245 ± 0.01087
Two r8	No DP	0.04628 ± 0.00459	0.04156 ± 0.00670	0.04552 ± 0.00250
Two r8	Clip only	0.04483 ± 0.00710	0.03795 ± 0.00832	0.04383 ± 0.00209
Two r8	8	0.04432 ± 0.00443	0.03764 ± 0.00727	0.04436 ± 0.00200
Two r8	4	0.04282 ± 0.00646	0.03991 ± 0.00452	0.04518 ± 0.00452
Two r8	2	0.03761 ± 0.00718	0.03276 ± 0.00489	0.04102 ± 0.00349
Two r8	1	0.01843 ± 0.00691	0.01794 ± 0.00611	0.03058 ± 0.00535

810 original rows aggregate into 54 unit/level settings. These diagnostic alternatives are not privacy-accounted serving mechanisms.

S5 Pulse/reset and replication diagnostics

S5.1 E2/E2b endpoints

Churn is the percentage of top-10 set entries that disagree, not the percentage of affected users or NDCG loss. A normalized RMS ratio can be large because the numerator grows or because the initial denominator is small. Reported means must be read alongside the seed-level and median records for unstable Two-factor cases. E2 uses all 942 users as observers; E3 below uses a separately selected sample of 128.

Table S12: E2/E2b lag-ten diagnostics across both pulse amplitudes and all reset branches.

Study/unit	Level	α	Branch	Churn (%)	Score RMS	RMS/pulse
E2 raw FixedB r8	1	0.25	local_reset	4.2197	2.098×10^{-2}	1.2105
E2 raw FixedB r8	1	0.25	retained	4.2378	2.108×10^{-2}	1.2164
E2 raw FixedB r8	1	0.25	shared_reset	0.0945	1.454×10^{-3}	0.0840
E2 raw FixedB r8	1	1.00	local_reset	14.8376	8.384×10^{-2}	1.2091
E2 raw FixedB r8	1	1.00	retained	14.9406	8.424×10^{-2}	1.2149
E2 raw FixedB r8	1	1.00	shared_reset	0.3747	5.825×10^{-3}	0.0841
E2 raw FixedB r8	2	0.25	local_reset	2.9013	1.220×10^{-2}	1.2609
E2 raw FixedB r8	2	0.25	retained	2.9140	1.223×10^{-2}	1.2641
E2 raw FixedB r8	2	0.25	shared_reset	0.0648	5.841×10^{-4}	0.0605
E2 raw FixedB r8	2	1.00	local_reset	10.4193	4.878×10^{-2}	1.2602
E2 raw FixedB r8	2	1.00	retained	10.4735	4.890×10^{-2}	1.2633
E2 raw FixedB r8	2	1.00	shared_reset	0.2038	2.337×10^{-3}	0.0606
E2 raw Full	1	0.25	local_reset	4.5191	3.273×10^{-2}	1.1772
E2 raw Full	1	0.25	retained	4.5425	3.295×10^{-2}	1.1850
E2 raw Full	1	0.25	shared_reset	0.1518	2.913×10^{-3}	0.1048
E2 raw Full	1	1.00	local_reset	15.5616	1.309×10^{-1}	1.1769
E2 raw Full	1	1.00	retained	15.6571	1.318×10^{-1}	1.1847
E2 raw Full	1	1.00	shared_reset	0.5212	1.170×10^{-2}	0.1052
E2 raw Full	2	0.25	local_reset	3.4214	1.759×10^{-2}	1.1944
E2 raw Full	2	0.25	retained	3.4299	1.766×10^{-2}	1.1991
E2 raw Full	2	0.25	shared_reset	0.0775	1.130×10^{-3}	0.0768
E2 raw Full	2	1.00	local_reset	11.9766	7.034×10^{-2}	1.1941
E2 raw Full	2	1.00	retained	12.0191	7.062×10^{-2}	1.1988
E2 raw Full	2	1.00	shared_reset	0.3206	4.531×10^{-3}	0.0769
E2 raw Two r8	1	0.25	local_reset	46.4713	2.107×10^{2}	4301.4699
E2 raw Two r8	1	0.25	retained	46.5403	2.107×10^{2}	4301.6017
E2 raw Two r8	1	0.25	shared_reset	11.1465	4.027×10^{1}	821.7067
E2 raw Two r8	1	1.00	local_reset	49.4352	1.864×10^{2}	953.4395
E2 raw Two r8	1	1.00	retained	49.7845	2.034×10^{2}	1040.4037
E2 raw Two r8	1	1.00	shared_reset	17.8546	1.783×10^{2}	911.1964
E2 raw Two r8	2	0.25	local_reset	18.2856	1.684×10^{-1}	11.6037
E2 raw Two r8	2	0.25	retained	18.3567	1.688×10^{-1}	11.6316
E2 raw Two r8	2	0.25	shared_reset	3.6019	3.321×10^{-2}	2.2633
E2 raw Two r8	2	1.00	local_reset	19.4331	1.883×10^{-1}	3.2189
E2 raw Two r8	2	1.00	retained	19.3461	1.867×10^{-1}	3.1951
E2 raw Two r8	2	1.00	shared_reset	9.3960	8.139×10^{-2}	1.4567
E2b aligned Two r8	1	0.25	local_reset	4.6274	1.892×10^{1}	386.1677
E2b aligned Two r8	1	0.25	retained	4.6423	1.892×10^{1}	386.1958
E2b aligned Two r8	1	0.25	shared_reset	0.1125	1.011×10^{-2}	0.2209

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Table S12 (continued)

Study/unit	Level	α	Branch	Churn (%)	Score RMS	RMS/pulse
E2b aligned Two r8	1	1.00	local_reset	15.8248	7.921×10^{1}	404.9841
E2b aligned Two r8	1	1.00	retained	15.8811	7.922×10^{1}	405.0528
E2b aligned Two r8	1	1.00	shared_reset	0.4204	4.117×10^{-2}	0.2252
E2b aligned Two r8	2	0.25	local_reset	2.1550	1.916×10^{-2}	1.2983
E2b aligned Two r8	2	0.25	retained	2.1561	1.918×10^{-2}	1.2995
E2b aligned Two r8	2	0.25	shared_reset	0.0287	5.870×10^{-4}	0.0404
E2b aligned Two r8	2	1.00	local_reset	7.3779	7.651×10^{-2}	1.2961
E2b aligned Two r8	2	1.00	retained	7.3949	7.658×10^{-2}	1.2972
E2b aligned Two r8	2	1.00	shared_reset	0.1083	2.367×10^{-3}	0.0407

Each entry averages two draws within seed, then five seeds. A mean ratio can be dominated by a small denominator or an unstable seed; original medians and seed tables remain available.

Table S13: Transferred MovieLens-1M validation NDCG@10.

Method	$\epsilon = 1$	$\epsilon = 2$
Full	0.019533 ± 0.004151	0.024624 ± 0.001777
FixedB r8	0.013661 ± 0.004574	0.016482 ± 0.004476
Two r8	0.015254 ± 0.005194	0.015440 ± 0.004567
DP popularity	0.026124 ± 0.000168	0.026161 ± 0.000104

Mean and seed SD, five seeds. Settings transfer without retuning. No ML-1M test scoring. Source: validation_summary.csv.

S5.2 E3 complete endpoint register

"None" disables balancing only in the diagnostic continuation from the same balanced checkpoint. Raw and aligned couplings change the joint comparison; removing balancing changes the continuation algorithm itself. Validation differences in this table concern only observer users. They are distinct from full-population ML-1M validation in the main report.

Table S14: All E3 shared-reset lag-ten settings.

Data	Unit	Level	Balance/coupling	Churn (%)	Score RMS	Δ val. NDCG
ml100k	Two r8	1	none/aligned	0.3438 ± 0.0642	1.100×10^{-2}	0.000000
ml100k	Two r8	1	none/raw	0.3438 ± 0.0642	1.100×10^{-2}	0.000000
ml100k	Two r8	1	svd/aligned	0.4219 ± 0.1278	1.106×10^{-2}	0.000040
ml100k	Two r8	1	svd/raw	18.6641 ± 3.6991	2.028×10^{-1}	0.001452
ml100k	Two r8	2	none/aligned	0.0781 ± 0.0276	2.245×10^{-3}	0.000027
ml100k	Two r8	2	none/raw	0.0781 ± 0.0276	2.245×10^{-3}	0.000027
ml100k	Two r8	2	svd/aligned	0.0859 ± 0.0327	2.272×10^{-3}	0.000226
ml100k	Two r8	2	svd/raw	5.2031 ± 5.1075	4.922×10^{-2}	0.000919
ml1m	FixedB r8	1	svd/raw	0.0859 ± 0.0579	2.625×10^{-4}	-0.000011
ml1m	FixedB r8	2	svd/raw	0.0938 ± 0.0810	1.252×10^{-4}	0.000014
ml1m	Full	1	svd/raw	0.1328 ± 0.0592	5.509×10^{-4}	0.000000
ml1m	Full	2	svd/raw	0.0625 ± 0.0524	2.243×10^{-4}	-0.000014
ml1m	Two r8	1	none/aligned	0.0703 ± 0.0642	2.056×10^{-4}	0.000000
ml1m	Two r8	1	none/raw	0.0703 ± 0.0642	2.056×10^{-4}	0.000000
ml1m	Two r8	1	svd/aligned	0.0625 ± 0.0761	2.039×10^{-4}	0.000018
ml1m	Two r8	1	svd/raw	5.2344 ± 4.1984	9.777×10^{-3}	-0.000340
ml1m	Two r8	2	none/aligned	0.0156 ± 0.0214	1.043×10^{-4}	0.000000
ml1m	Two r8	2	none/raw	0.0156 ± 0.0214	1.043×10^{-4}	0.000000
ml1m	Two r8	2	svd/aligned	0.0078 ± 0.0175	9.923×10^{-5}	0.000000
ml1m	Two r8	2	svd/raw	3.3438 ± 2.5386	5.122×10^{-3}	0.000610

Churn includes seed SD; other columns show seed means. Validation differences use only 128 observer users and have limited resolution. Starting checkpoint epsilon does not account for continuation.

S6 Ranking-tail screening

The exact conditional law is established mathematics, numerically evaluated for the prescribed states. The advancement gate requires matched-Gaussian error above 0.01 on at least 5% of primary pairs in both datasets. No primary pair exceeds this threshold. The table also retains the linearized approximation as a secondary diagnostic.

Table S15: Complete E4 state-group screening summary.

Data	Unit	Level	Matched max error	Linear max error	Mean cross (%)
ml100k	Two r16	1	6.637×10^{-6}	3.481×10^{-5}	0.03166
ml100k	Two r16	2	4.163×10^{-6}	2.862×10^{-5}	0.02242
ml100k	Two r32	1	7.669×10^{-6}	7.262×10^{-5}	0.07498
ml100k	Two r32	2	5.413×10^{-6}	5.551×10^{-5}	0.05992
ml100k	Two r4	1	5.851×10^{-6}	5.089×10^{-6}	0.00701
ml100k	Two r4	2	3.687×10^{-6}	3.550×10^{-6}	0.00464
ml100k	Two r8	1	6.515×10^{-6}	1.303×10^{-5}	0.01526
ml100k	Two r8	2	3.790×10^{-6}	1.163×10^{-5}	0.00959
ml1m	Two r8	1	5.508×10^{-6}	3.397×10^{-5}	0.00253
ml1m	Two r8	2	3.545×10^{-6}	1.870×10^{-5}	0.00105

Each row summarizes 320 prescribed records. Cross is pairwise margin-variance share, not E1 whole-matrix energy. Maximum error is an absolute probability difference.

S7 Calibration, selection, and communication

Table S16: Calibration of the iterative and one-release mechanisms.

Mechanism	ϵ target	σ	Achieved ϵ	q	T
training	8	0.805072021	7.967513	0.1	50
popularity	8	0.600229072	8.000000	1.0	1
training	4	1.151928711	3.963885	0.1	50
popularity	4	1.081161850	4.000000	1.0	1
training	2	1.760449219	1.993278	0.1	50
popularity	2	1.993812446	2.000000	1.0	1
training	1	2.977490234	0.990062	0.1	50
popularity	1	3.730631635	1.000000	1.0	1

Delta is 10^{-5} throughout. Popularity is a full-population single Gaussian release; its multiplier is not interchangeable with collaborative calibration.

Table S17: Validation-only rank selections for E1.

Method	Level	Selected rank	Validation NDCG@10
Two	No DP	32	0.048192
Two	Clip only	32	0.049694
Two	8	32	0.049215
Two	4	32	0.045888
Two	2	8	0.037609

Continued on next page

Table S17 (continued)

Method	Level	Selected rank	Validation NDCG@10
Two	1	4	0.021543
FixedB	No DP	32	0.053402
FixedB	Clip only	32	0.055806
FixedB	8	32	0.048997
FixedB	4	32	0.039215
FixedB	2	16	0.027423
FixedB	1	16	0.011983

Table S18: Modeled dense tensor communication for retained historical methods.

Model	Bytes/client/round	Rounds to best	GB to best	GB complete
B1	861,184	1158.0	93.911	174.946
B3 r4	55,872	2984.0	15.704	20.971
B3 r8	111,744	3064.0	32.263	42.790
B3 r16	223,488	3676.0	77.388	98.415
B3 r32	446,976	4284.0	180.329	222.297
B2	861,184	50.0	4.026	4.026
B4 r4	55,872	50.0	0.261	0.261
B4 r8	111,744	50.0	0.522	0.522
B4 r16	223,488	50.0	1.045	1.045
B4 r32	446,976	50.0	2.090	2.090

Decimal GB; realized-participation totals averaged over seeds. Headers, secure aggregation, compression, and device costs excluded. Source: communication.csv.

S8 Additional figures

These figures accompany results stated in the main report.

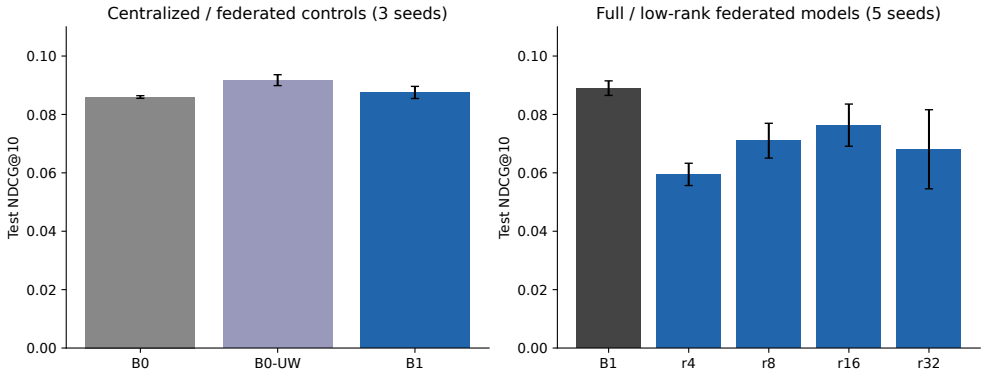


Figure 1: Left: matched three-seed centralized and federated test NDCG@10. Right: five-seed converged B1 and B3. Whiskers show seed SD. The panels have different seed sets. Overall weighting-decomposition intervals include zero; all four adjusted B3-versus-B1 intervals are negative. Sources: retained centralized control and final five-seed tables.

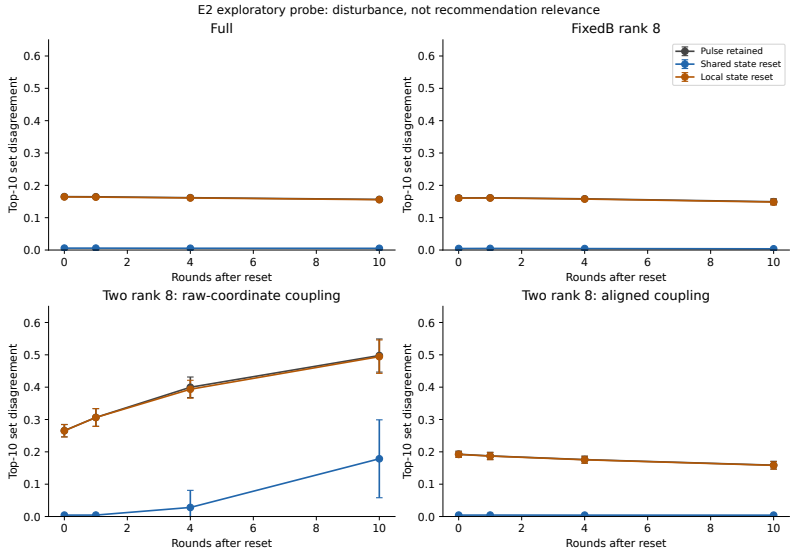


Figure 2: E2/E2b top-10 disagreement for retained pulse, shared reset, and local reset at starting target epsilon one and pulse amplitude one. Means and SDs use five checkpoint seeds after averaging two draws within seed. Candidate sets exclude training positives. The interventions access private states and measure prediction disagreement, not relevance or accuracy gains.

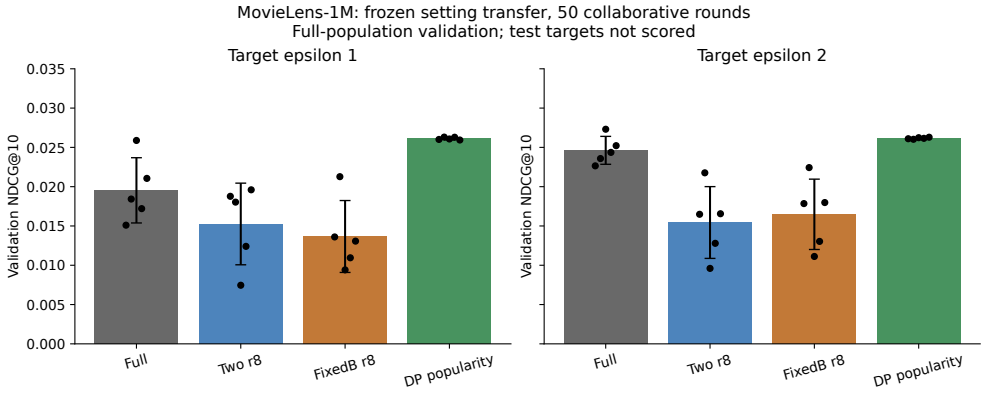


Figure 3: Full-population ML-1M validation NDCG@10 under frozen transferred settings. Bars and whiskers give five-seed means and SDs; dots show each seed. Collaborative models use 50 rounds and popularity uses one separately calibrated release. Point differences are descriptive. ML-1M test targets have not been scored.

S9 Detailed experimental record

S9.1 Protocol detail

On ML-100K, 316 users have a validation timestamp equal to their final training timestamp, and 305 have equal validation and test timestamps; the fixed convention is therefore consequential and documented.

The retained ML-100K users have a mean of 56.7845 and median of 37.5 training positives, with a range of 1–376. Activity groups are defined solely from these counts: low activity has at most 22 positives (319 users), medium activity 23–61 (310 users), and high activity at least 62 (313 users). Six validation and fourteen test targets are cold items with no training-positive observation across the population. They remain in the principal evaluation.

S9.2 Diagnostic random streams and identity checks

E2 pulse streams use the tuple (training seed, 6702, pulse replicate). Future client, negative, and Gaussian streams use identifiers 6703, 6704, and 6705 respectively. Common streams are retained across branches and pulse amplitudes. E3 uses separately declared dataset-coded streams with identifiers 6710–6713 and observer selection identifier 6708. E4 uses its own fixed observer selection. Exact seed tuples are part of the frozen protocols and saved records, so a report rerender does not generate new diagnostic observations.

Synthetic checks compare zero-pulse branches with their references and both-reset branches over continued rounds. Shared resets enforce identical Q; local resets enforce identical P. The score decomposition includes all cross inner products. Alignment checks measure relative product error instead of assuming exact equality in float32. E4 checks moment formulas within Monte Carlo uncertainty and audits quadrature diagnostics separately. These checks support the implemented experiments; they do not prove an unrestricted mathematical correctness claim for every numerical state.

S9.3 Numerical environment

The pinned research environment records NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.3, PyTorch 2.14.0+cpu, Opacus 1.6.0, matplotlib 3.10.9, and the remaining versions in the pinned requirements file. Opacus is used for accountants, not a PrivacyEngine wrapping these models. Training and diagnostic observer precision are distinguished: model updates retain float32, while relevant diagnostic norms and closures use float64. E1 enforces one numerical worker thread; older centralized runs used their recorded CPU/thread configuration.

A fresh-environment historical B0 reproduction was identical to reported metric precision but not byte-identical for seed 42. Relative weight difference was approximately 7.8×10^{-6}, and one validation

rank changed from 596 to 595 without changing reported six-decimal metrics. A floating-point kernel explanation was plausible but unverified. Accordingly the report promises recorded seeds, environments, artifacts, and numerical audits, not universal bitwise identity on different systems.

S9.4 Further derivation notes

Separate factor averaging has an additional distinction. With client means $\bar{A}, \bar{B}$,

$$\frac{1}{K}\sum_{u=1}^{K} A_u B_u = \bar{A}\bar{B} + \frac{1}{K}\sum_{u=1}^{K}(A_u - \bar{A})(B_u - \bar{B}). \tag{1}$$

Thus averaging factors and averaging products generally differ by a cross-client covariance term. FixedB removes this discrepancy, but E1 does not independently intervene on it. It cannot be isolated as the reason for a utility difference from the existing comparisons.

The full shock is not generally Gaussian. For clean nonzero margin m, Chebyshev's inequality supplies the weak bound

$$\Pr(\text{sign flip}) \le \Pr(|\text{shock}| \ge |m|) \le \min(1, V_{ij}/m^2). \tag{2}$$

A variance alone does not justify a Gaussian tail formula for Two.

For scores, the corresponding minimizer satisfies $c^4 = M \left\|PB^T\right\|_F^2 / (\|A\|_F^2 \|P\|_F^2)$. An equal-factor-norm convention need not minimize either rectangular or user-weighted expression. The bilinear τ^4 term is unchanged by this scalar gauge transformation.

S9.5 Hypothesis ledger

Fewer shared noisy coordinates reliably improve recommendation.
Rejected as a general statement. Four favorable historical private contrasts coexist with weak-privacy losses, nonprivate losses, and a strong private popularity comparator.

Fixing one public factor is a sufficient new solution.
Unsupported by E1 utility. The implemented method is a controlled baseline.

The factor-noise product dominates the observed failures.
Not supported by its small measured matrix-energy share. E4 additionally rejects large Gaussian tail-risk approximation error in the prescribed states.

Noise causes a large general persistent-personalization failure.
The broad reading of raw Two-r8 distances is rejected by alignment and no-balancing controls. Modest local-state effects remain.

Factor-basis changes can dominate a paired noise-attribution measurement.
Supported within the declared simulator, counterfactual, datasets, and observation rules. Its external consequence remains open.

Public metadata or private ALS has already solved the remaining problem here.
Not tested; these remain future comparisons.

S9.6 Chronology of material revisions

Table S19: Material research iterations and their consequences.

Stage	What changed or was learned	Consequence for interpretation
Phase 0, 28 Sep.	Positive threshold, per-user chronology, hashed timestamp ties, candidate filtering, and raw/split hashes were fixed.	A common full-ranking protocol replaces inconsistent or sampled-candidate comparisons.
B0, 28–29 Sep.	Short patience and epoch caps could stop a still-improving configuration; validation-only convergence checks revised selection.	Earlier B0 numbers remain historical, not the final common-protocol reference.
B0/B1, 2 Oct.	A common longer stopping protocol and centralized B0-UW control were added.	User weighting partly explains the apparent federation difference; overall paired intervals include zero.
B2, 2 Oct.	T1000/C1.5/server2 was replaced after validation studies by T50/C1/server1.	Three design choices changed; a single-factor causal story about horizon is not supported.
B3	Initial rates caused divergence; a rank-32 convergence/stability gate required bounded correction before test scoring.	Report final rates and the one declared 16,000-round diagnostic ceiling. Do not silently drop failed attempts.
B4	Inherited long-horizon rates undertrained at 50 rounds; expanded searches and a bounded fallback selected finite settings.	Historical B4 has a larger tuning budget than B2, limiting a dimension-only comparison.
E1, 2 Oct.	A new equal-four-candidate extension matched factorized initializations and added DP popularity and clipping controls.	E1 is separately named and frozen; it does not overwrite B4. Eleven search failures remain counted.
E1 diagnostics	Float32 local norm summaries overflowed for a finite trajectory; float64 analysis corrected the summary.	Training, checkpoint contents, and test ranks are unchanged. Extreme finite results stay visible.
E1 verification	Live test stdout written under a guarded results tree interfered with integrity fixtures. A rerun wrote to a temporary location.	The recorded passing run has 203 tests and 26 warnings; both logs are retained. The failure was a harness/output-location issue.
E2, 3 Oct.	120 declared pulse/reset probes suggested local memory and large Two-r8 distances.	Exploratory diagnosis, no relevance scoring or new DP claim.
E2b, 3 Oct.	An alignment control was declared after E2's large two-factor distances.	Adaptive follow-up sharply narrows the original interpretation.
E3, 3 Oct.	30 ML-1M training runs and 200 raw/aligned, balanced/unbalanced probes completed.	The large coupling effect depends on balancing; ML-1M validation is exposed, while its test is unscored.
E4, 3 Oct.	50 states and 3,200 prescribed pairs tested a known non-Gaussian margin law.	The practical-error gate failed. No post hoc expansion was used to rescue the candidate.

Table S19 (continued)

Stage	What changed or was learned	Consequence for interpretation
Report, 5 Oct.	Saved evidence was organized into LaTeX narrative, vector figures, generated tables, and supplementary material.	No new training, model selection, or test scoring was needed for report preparation.

S10 Source mapping and verification

Printed table family	Retained source under the submission's `data/` directory
Historical utility	`baselines/final_5seed_means.csv`
Historical contrasts	`baselines/paired_contrasts.csv`
E1 utility / comparisons	`effective_noise/test_means.csv`; `effective_noise/contrasts.csv`
Geometry	`effective_noise/geometry_means.csv`
Groups / local-vector ablations	`effective_noise/subgroups_test.csv`; `effective_noise/personalization_validation.csv`
E2 / E2b	`noise_memory/summary.csv`; `noise_memory_aligned/summary.csv`
E3 endpoints / validation	`noise_replication/endpoint_summary.csv`; `noise_replication/validation_summary.csv`
E4 screen	`ranking_tails/screen_summary.csv`
Calibration / E1 rank selection	`effective_noise/accounting.csv`; `effective_noise/rank_selection.csv`
Communication	`baselines/communication.csv`

The script `scripts/generate_tables.py` formats these inputs and records their SHA-256 hashes in `quality/table_provenance.json`. The surrounding CS-670 bundle contains seed tables, freeze records, protocols, the bibliography, and the source provenance record. The archive verification script checks original evidence without running models. Tables may be rebuilt from the archived CSVs; editing generated numerical cells by hand would break their provenance relationship.